

Feature Transport Across Grammatical Roles in Pythia-410M

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Abstract

Cross-position information flow in transformers has been studied through subject–verb agreement [Hao and Linzen, 2023], pronoun resolution circuits [Wang et al., 2023], binding subspaces [Feng and Steinhardt, 2023], OV singular-vector analysis [Ahmad et al., 2025], and activation transport operators [Szablewski and Masiak, 2025]. We measure one specific mechanism—a low-rank linear map from a source token’s representation to a consuming position—across three grammatical roles in Pythia-410M-deduped. Object channels (centered R^2 0.52–0.56) are stronger than the subject channel (0.41), despite English having no object agreement. Standard grammatical features (number, gender, animacy) account for 27% of the subject channel’s variance; unsupervised clustering of the remaining content recovers 7 semantic axes (place, body-part, food, person, animal, household object, water body), each of which causally shifts next-token predictions when patched, while matched random-direction controls do not. The transported copy is decodable at the consuming position from layer 8 but only becomes causal at layers 20–23. All results are from one model and one language (English). Code and data: <https://github.com/EvidentSolutions/Deposit>.

1 Introduction

Cross-position information flow in transformers has been studied through several phenomena. Wang et al. [2023] identified the circuit by which GPT-2 resolves indirect object pronouns. Hao and Linzen [2023] showed that subject–verb agreement in BERT relies on a linear encoding of subject number that can be read at the verb position. Feng and Steinhardt [2023] found that binding in Pythia operates through low-rank subspaces. Ahmad et al. [2025] decomposed OV circuits into singular-vector directions that control gender pronoun prediction. Szablewski and Masiak [2025] proposed rank-constrained activation transport operators for tracing feature flow through the residual stream.

Most of this work examines one grammatical role at a time, typically the subject in agreement constructions. The low-rank linear map framing—fitting a map from a source token’s representation to the consuming position and testing what features it carries—has been applied mainly to agreement and binding. We apply the same measurement to three grammatical roles in Pythia-410M-deduped (EleutherAI), including roles where no morphological constraint requires cross-position transport. In principle, any position can transport features to any other

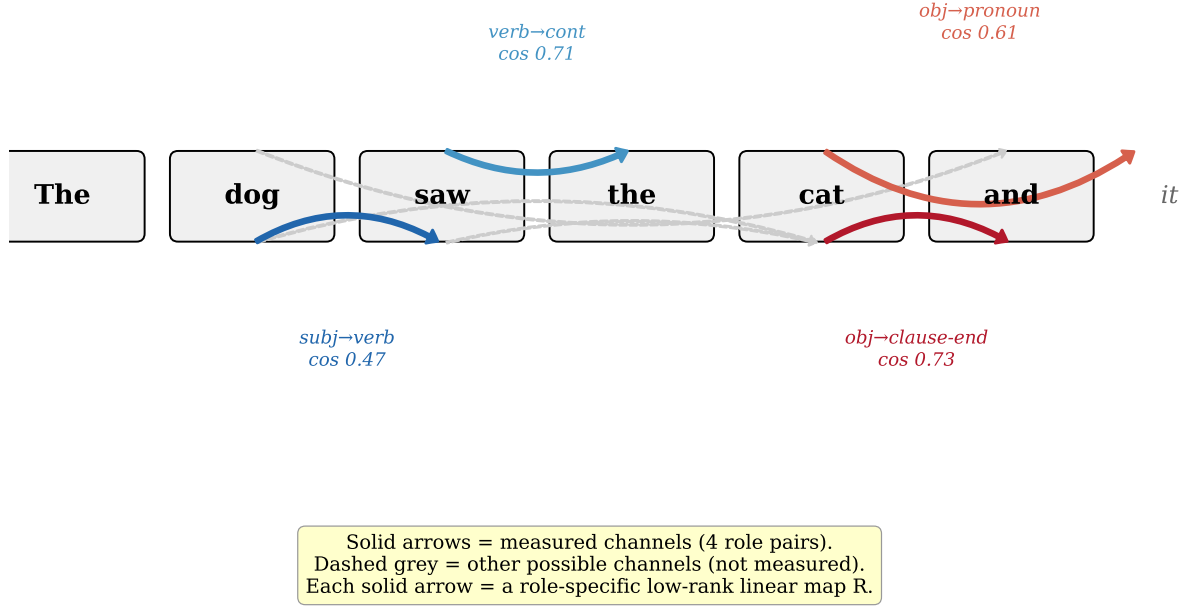


Figure 1: Deposit channels across grammatical roles (illustrative sentence). Solid coloured arrows show source–consumer pairs; dashed grey arrows indicate other possible channels that were not measured. Three noun roles (subject→verb, object→pronoun, object→clause-end) are quantified in Table 1; verb→continuation is illustrated but not included in the centered R^2 analysis (it uses a different feature battery).

position via attention; we measure three specific source–consumer pairs out of many possible ones, and make no claim about pairs we did not test.

We report three sets of measurements:

1. **Deposit channel strength by role.** Centered R^2 of a ridge-fit linear map for subject→verb, object→pronoun, and object→clause-end. Object channels are stronger than the subject channel.
2. **Channel content.** Standard grammatical features (number, gender, animacy) explain 27% of the channel’s variance. Unsupervised clustering of the remaining content, followed by per-cluster causal intervention, recovers 7 additional semantic axes, all causally validated.
3. **Timing of causal use.** The transported copy is decodable at the consuming position from layer 8 but has no measurable causal effect on the prediction until layer 16, reaching full effect at layers 20–23.

2 Method

Model. All experiments use Pythia-410M-deduped [Biderman et al., 2023], a 24-layer autoregressive transformer with $d_{\text{model}} = 1024$.

2.1 Deposit map

We define a *deposit map* as a linear map $R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ from the source token’s residual-stream representation at a lexical layer to the consuming position’s representation at a transport layer. We use layer 8 as the lexical layer and layer 12 as the transport layer throughout. A grid search over source layers $\{2, 4, \dots, 16\}$ and consumer layers $\{4, 6, \dots, 22\}$ (Figure 5) shows that the centered R^2 forms a broad plateau: any source layer ≤ 10 paired with any consumer layer in $\{12, \dots, 18\}$ gives $R^2 \approx 0.33\text{--}0.38$. L8→L12 ($R^2 = 0.37$) is near the maximum (L8→L14, $R^2 = 0.38$). Below consumer layer 10, the transport signal is weak ($R^2 < 0.23$); the source layer matters little once it exceeds 2.

Given N source words, let $X \in \mathbb{R}^{N \times d}$ be the centered lexical representations (source token, layer 8) and $Y \in \mathbb{R}^{N \times d}$ the centered transport representations (consuming position, layer 12). We fit R by ridge regression:

$$R = (X^\top X + \lambda I)^{-1} X^\top Y, \quad \lambda = 0.1 \cdot \frac{1}{d} \sum_j \|X_{:,j}\|^2. \quad (1)$$

We evaluate on held-out words (75/25 train/test split, randomized over 10 random splits) by centered R^2 : R^2 computed on mean-subtracted representations, isolating the word-specific component from global positional structure (see §3.1). We also report R^2 restricted to the 30-dimensional channel subspace (top-30 singular vectors of R) and cross-category controls where R is fit on wrong-word pairings (e.g., dog’s source $\rightarrow$ chair’s consumer representation).

2.2 Feature battery

For each binary feature (number: singular/plural; gender: masculine/feminine; animacy: animate/inanimate), we compute the feature direction in lexical space as the difference of class means, and similarly in transport space. We test whether R maps one to the other by measuring $\cos(R \cdot \ell_F, t_F)$, where ℓ_F and t_F are the lexical and transport feature directions respectively.

Identity stripping is measured by η^2 of word identity (one-way ANOVA on word labels) in the channel coordinates: $\eta^2 = 1.0$ means every word is individually distinguishable; $\eta^2 \approx 0$ means the channel carries only abstract features.

2.3 Discover-validate sweep

1. Extract the 30-dimensional input channel of R (top-30 left singular vectors of R).
2. Cluster words in this space using k -means ($k = 8$).
3. For each cluster C , compute a cluster direction u_C (mean of cluster members minus global mean, in channel coordinates). Measure transport cosine on held-out words: $\cos(R \cdot u_C^{\text{lex}}, u_C^{\text{tr}})$.
4. Causal test: in the frame “The $\langle N \rangle$ is _____”, patch the component of the noun’s residual along u_C to the cluster mean (vs. the complement mean). Measure the shift in a learned category readout (mean logits for cluster-typical continuations minus global mean logits). Compare to a matched random-direction control (same norm, same projection magnitude,

Table 1: Deposit map quality by grammatical role (centered R^2 , 10 random splits).

Role	Centered R^2	Cross-cat R^2	ΔR^2	Chan. R^2	η^2	Eff. rank
Subject $\rightarrow$ verb	0.41	-0.42	0.83	0.94	0.16	32
Object $\rightarrow$ pronoun	0.52	-0.33	0.85	0.96	0.30	33
Object $\rightarrow$ clause-end	0.56	-0.65	1.22	0.95	0.55	36

Centered R^2 : ridge regression on mean-subtracted (word-specific) representations, 75/25 split, averaged over 10 random splits. Cross-cat R^2 : same map fitted on cross-category word pairings (dog $\rightarrow$ chair); negative values mean the wrong-word map hurts prediction. ΔR^2 = matched - cross-category. Chan. R^2 : R^2 restricted to the 30-dim channel subspace (top-30 singular vectors of R). η^2 = ANOVA on word identity (1.0 = identity preserved). Verb $\rightarrow$ continuation omitted (different feature battery; see text).

random orientation). A cluster is validated if causal effect > 0.3 and $> 3\times$ the random control’s effect.

2.4 Layer-swept causal test

To test whether the *transported copy* (not the source) is causal, we use the frame “The $\langle N \rangle$ near the old table is a type of ____”. The source noun is at position 1; the consuming position is the final token (7+ positions away, past a distractor noun “table”). We patch a discovered axis (place vs. object category) at the consuming position only, at each layer from L4 to L23, and measure the shift in a readout defined as $\text{logsumexp}(\text{logits}[\text{place words}]) - \text{logsumexp}(\text{logits}[\text{object words}])$, where place words = {city, town, place, country, area, region} and object words = {object, thing, tool, device, item, material}. We compare to random-direction patches at the same position and layer. We use 14 place nouns and 14 object nouns.

3 Results: role comparison

3.1 Three channels measured

We measured deposit maps for three source-consumer pairs. Stimuli follow the templates:

- **Subject $\rightarrow$ verb:** lexical = “The $\langle w \rangle$ near the table” (w at position 1); transport = verb position in “The $\langle w \rangle$ near the $\langle N2 \rangle$ $\langle V \rangle$ ” (position 5).
- **Object $\rightarrow$ pronoun:** lexical = “The $\langle S \rangle$ saw the $\langle w \rangle$ by the wall” (w at position 4); transport = “it” in “The $\langle S \rangle$ saw the $\langle w \rangle$ near the $\langle N2 \rangle$ and it” (position 9).
- **Object $\rightarrow$ clause-end:** same lexical frame; transport = “wall” in “The $\langle S \rangle$ saw the $\langle w \rangle$ by the wall” (position 7).

Noun vocabulary: 112 words across 7 semantic categories (animal, person, object, place, abstract, food, body-part), filtered to single-token items. 75/25 train/test split, averaged over 10 random splits.

Table 1 shows the results using centered R^2 (word-specific component only; see §3.1 for why uncentered cosine is not appropriate). Object channels have higher centered R^2 than the subject channel: object $\rightarrow$ clause-end (0.56) and object $\rightarrow$ pronoun (0.52) vs. subject $\rightarrow$ verb (0.41). English has no morphological object agreement; the transport exists without morphological pressure.

All cross-category R^2 values are strongly negative (-0.33 to -0.65), confirming that the map captures word-specific content, not global structure.

Within the 30-dimensional channel subspace (top-30 singular vectors of R), R^2 rises to 0.94–0.96 for all roles. The “low” full-space R^2 (0.41–0.56) reflects the channel’s low dimensionality relative to the 1024-dim residual stream, not weak transport.

Identity stripping varies by role. Subject→verb strips most aggressively ($\eta^2 = 0.16$): 84% of word-identity information is removed, leaving only abstract features. Object→clause-end preserves more identity ($\eta^2 = 0.55$), consistent with the specific object constraining the continuation more than the specific noun constrains agreement.

Cross-role transfer fails: R_{subj} applied to object data gives centered R^2 near zero. The maps are role-specific instantiations of the same architectural mechanism.

Global-structure control. The uncentered held-out cosine (used in prior work) conflates word-specific transport with global residual-stream structure shared between positions. To separate these, we ran two controls using 74 words across 7 semantic categories (10 random splits each).

Uncentered cosine (used in prior work on deposit maps) is dominated by global structure: matched = 0.477, cross-category swap (dog→chair, king→river) = 0.474, fixed-word (all sentences contain “dog”) = 0.504. The word-specific increment in this metric is near zero.

Centered R^2 (mean-subtracted across words, isolating the word-varying component) tells a different story: matched $R^2 = 0.43$, cross-category $R^2 = -0.30$ (a map fit on wrong pairings actively hurts prediction of the word-specific component). The word-specific ΔR^2 is 0.73. This holds across context lengths: with a 25-token filler prefix, $\Delta R^2 = 0.85$; with a 45-token prefix, $\Delta R^2 = 0.69$.

The deposit map captures genuine word-specific transport, but the uncentered held-out cosine is not an appropriate metric for measuring it—it is swamped by the ~ 1000 -dimensional global structure. The feature battery (Table 2) and the causal validation (§4, §5) are the appropriate evidence. The centered R^2 of the map (0.43) is a better measure of word-specific transport quality, though it too is influenced by the choice of regularisation and train/test split.

3.2 Feature re-encoding through one map

A single ridge-fit map R (not fitted per feature) reproduces the transport of all three tested feature directions:

Table 2: Per-feature transport cosine $\cos(R \cdot \ell_F, t_F)$ by role. The map R is fit per role on word-level representations (not per feature). Raw cos = cosine between lexical and transport feature directions without the map.

Feature	Subj→verb	Obj→pronoun	Obj→clause-end
Number	0.77 (raw 0.23)	0.93	0.83
Gender	0.80 (raw 0.30)	0.75	0.76
Animacy	0.94 (raw 0.14)	0.97	0.93

Raw cosines (without the map) shown for subject→verb only; comparable values for the other roles.

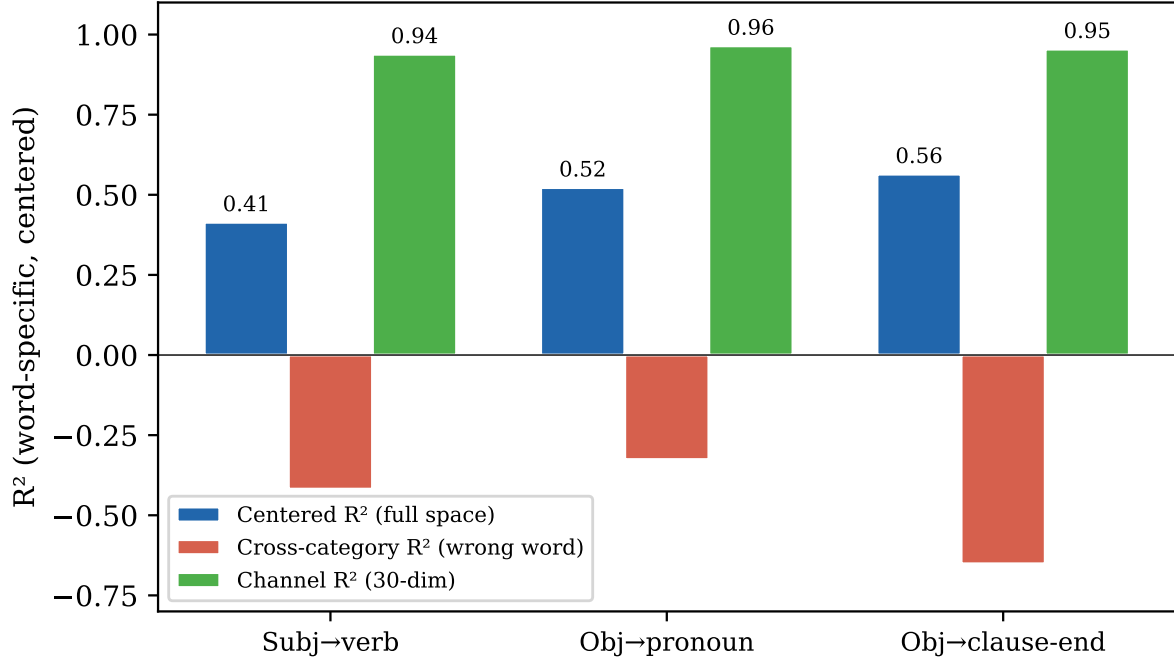


Figure 2: Word-specific transport quality by role. Blue: centered R^2 on the full 1024-dim space. Red: cross-category control (map fit on wrong-word pairings; negative values confirm word-specificity). Green: R^2 restricted to the 30-dim channel subspace (top-30 singular vectors of R).

All three noun-role channels transport all three features (cosine 0.75–0.97). Without the map, the cosines are low (0.14–0.30 for subject→verb); the map raises them substantially. The map has effective rank ≈ 30 out of 1024 dimensions: a low-rank projection that re-encodes prediction-relevant features while discarding lexical identity.

4 Results: channel content beyond grammatical features

4.1 Named features cover 27%

We projected the lexical representations of 81 nouns (spanning 7 coarse semantic categories: person, animal, object, abstract, place, body-part, food) into the 30-dimensional input channel of R (top-30 left singular vectors). Standard grammatical features (number, gender, animacy, person-vs-animal, concrete-vs-abstract) explain 27% of the channel’s variance. The remaining 73% carries structure not captured by these features.

4.2 Unsupervised clustering

k -means clustering ($k = 8$) on the channel-projected words recovers semantic groupings (Table 3). We validated each cluster using the discover-validate procedure described in §2.3.

One cluster slot (cluster 3) was empty after k -means. All 7 non-empty clusters both transport through R (held-out cosine 0.76–0.93) and shift the next-token distribution when patched, while the matched random-direction control does not.

Table 3: Discovered semantic axes: transport quality and causal validation.

Cluster	Example words	n	Transport cos	Causal effect	Random effect
Food	bread, seed, apple, rice	17	0.83	79.8	−0.07
Person	teacher, actor, king	16	0.93	139.0	0.02
Body/abstr.	heart, bone, faith, head	35	0.91	61.2	−0.02
Household	plate, cup, box, desk	10	—	60.9	0.01
Animal	dog, frog, deer, wolf	19	0.76	89.8	0.13
Place	country, harbor, forest	15	0.85	96.6	0.01
Water	lake, river, grass	3	—	103.2	−0.11

Transport cos: $\cos(R \cdot u_C^{\text{lex}}, u_C^{\text{tr}})$ on held-out words. Causal effect: shift in category readout when patching the cluster axis at the noun position. Random effect: same measurement with a matched random-direction control. “—” = too few held-out words for reliable transport estimate. k -means with $k = 8$ produced 7 non-empty clusters; all 7 pass validation (causal effect > 0.3 and $> 3 \times$ random).

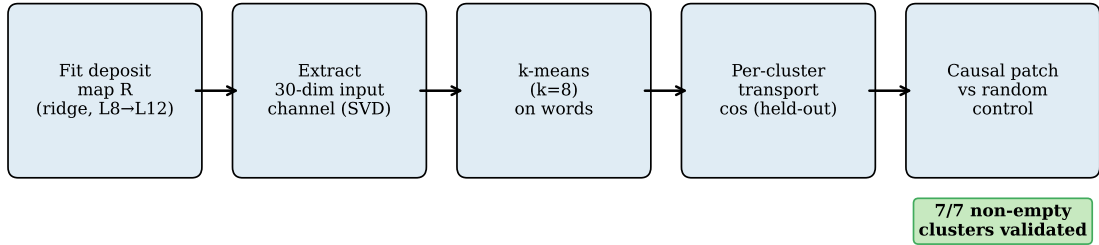


Figure 3: The discover–validate pipeline. Words are projected into the 30-dimensional input channel of R (via SVD), clustered with k -means, and each cluster is tested for (a) transport through R on held-out words and (b) causal effect on next-token predictions vs. a matched random-direction control. All 7 non-empty clusters passed validation.

4.3 Detailed validation: place and body-part axes

Two discovered axes were examined with separate causal tests using dedicated stimulus frames and readout word sets:

- **Place axis** (frame: “The $\langle N \rangle$ is a type of _____”): transport cosine 0.85, causal effect +4.16 on the place-vs-object readout, random control −0.002.
- **Body-part axis** (frame: “The $\langle N \rangle$ is a part of the _____”): transport cosine 0.89, causal effect +2.76, random −0.005.

These transport cosines are comparable to the hand-named animacy axis (~ 0.94). The channel carries semantic category information at the same fidelity as grammatical features.

5 Results: the transported copy becomes causal late

The transported copy is decodable at the consuming position from layer 8 (probe accuracy ≈ 1.0 from the role deposit tests). To measure when it becomes *causal*, we patched the place-axis

Table 4: Causal effect of patching the place axis at the consuming position, by layer.

Layer	Causal effect	Random effect
L4	0.05	≈ 0
L8	0.10	0.001
L12	0.82	-0.003
L16	2.19	-0.001
L20	4.28	-0.003
L23	4.48	-0.004
Source noun (L8)	3.19	≈ 0

Causal effect: shift in $\text{logsumexp}(\text{logits}[\text{place words}]) - \text{logsumexp}(\text{logits}[\text{object words}])$ when patching the place-axis component to the place-class mean vs. the object-class mean. Random effect: same measurement with a matched random direction. Source noun row: patching the place axis at the noun’s own position at L8.

component at the consuming position (final token in “The $\langle N \rangle$ near the old table is a type of _____”) at each layer from L4 to L23. The source noun is at position 1; only the transported copy exists at the consuming position.

The causal effect is near zero at L8 where the feature is decodable, and reaches its maximum at L20–23—the layers where the model writes the prediction (Table 4). The source noun’s features are causal early (effect 3.19 at L8), confirming the information exists at the source; the model reads the transported copy only at the late write-out.

A single-layer causal test at L8 (where the feature is readable) would find no effect and conclude the transported copy is not used. The layer sweep shows it *is* used, but later (Figure 4).

6 Related work

OV abstraction and low-rank structure. Elhage et al. [2021] described OV matrices as low-rank linear maps in their mathematical framework for transformer circuits. Xiao et al. [2024] showed that an OV circuit can transform an object token to its superclass (Appendix G.1). Feucht et al. [2025] demonstrated that concept-induction heads strip token identity while preserving abstract attributes. The deposit map measured here is an instance of this known structure. What we add is the cross-role comparison and the channel content analysis.

Pronoun resolution. The indirect object identification circuit [Wang et al., 2023] is the most-studied cross-position circuit in mechanistic interpretability. It is framed as circuit discovery rather than feature transport, but studies the same underlying phenomenon. Dunefsky and Cohan [2024] used pronoun prediction as a test case for feature vectors.

Binding subspaces. Feng and Steinhardt [2023] found that binding in Pythia operates through low-rank subspaces. Dai et al. [2024] localized Ordering ID representations that causally determine binding behavior. We measure the same kind of mechanism for additional roles (object, verb) and find that maps are role-specific (applying one role’s map to another’s data fails).

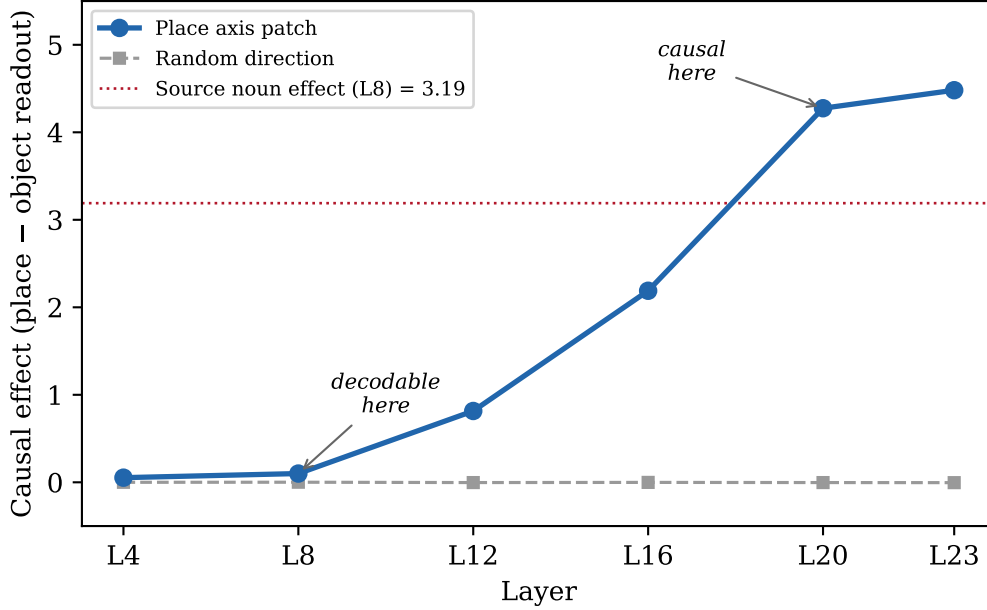


Figure 4: Causal effect of patching the place axis at the consuming position, by layer. The feature is decodable from L8 but has near-zero causal effect until L16, reaching full effect at L20–23. The dotted line shows the effect of patching the source noun directly at L8. The random-direction control (dashed grey) is near zero throughout.

OV singular vectors. Ahmad et al. [2025] decomposed OV circuits into singular-vector directions that control gender pronoun prediction, showing that individual heads implement multiple overlapping subfunctions aligned with distinct singular directions.

Activation transport operators. Szablewski and Masiak [2025] proposed rank-constrained linear maps for tracing feature transport through the residual stream. Their framework is general; our measurement is role-specific and includes causal validation of discovered channel content.

Agreement probing. Hao and Linzen [2023] traced subject number transport in BERT. Hernandez et al. [2024] studied factual associations as linear relational embeddings. Our measurements extend to roles without morphological agreement.

Causal probing. Activation patching [Meng et al., 2022] and automated circuit discovery [Conmy et al., 2023] are standard tools. The layer sweep in §5 shows that a single-layer causal test can miss a late-used transported signal.

Feature discovery. Engels et al. [2024] characterized SAE reconstruction error as mostly linear. Our discover-validate sweep uses unsupervised clustering on the deposit channel’s input space rather than SAE decomposition.

7 Discussion

Summary of findings. In Pythia-410M, the low-rank linear map that transports features from source tokens to consuming positions exists for objects, not only for subject–verb agreement. Object channels have higher centered R^2 than the subject channel (0.52–0.56 vs. 0.41). The subject channel carries features that are mostly grammatical (feature-subspace $R^2 = 0.86$); the object channel carries richer content (feature-subspace $R^2 = 0.43$), consistent with the object’s semantic identity mattering for prediction beyond agreement features. The channel carries semantic category information beyond standard grammatical features, and the transported copy becomes causal only at the model’s late layers.

Limitations. All results are from one model (Pythia-410M-deduped) and one language (English). We do not know whether these patterns hold in other architectures or at other scales. The deposit map is fit by ridge regression on a vocabulary of 112 nouns for the role comparison and 81 nouns for the channel content analysis; a larger or more diverse vocabulary might change the effective rank or identity-stripping measurements.

The linear map R captures only the linear component of cross-position transport. Any nonlinear composition by attention heads before writing would be missed. We extract the top-30 singular vectors of R for the channel content analysis; low-variance directions outside this subspace could carry additional features. The feature battery (5 hand-specified features plus 7 discovered clusters from k -means with $k = 8$) does not exhaustively characterize the channel’s content. The layer-pair sweep (Figure 5) shows that L8→L12 is near-optimal and the result is robust across a broad range of layer pairs, but the channel content analysis and causal tests were only run at this one pair.

We do not trace which attention heads compute the deposit or how the QK routing selects the source token. The causal tests measure shifts in category-level predictions (e.g., “place words become more probable”); we do not trace downstream effects on full generation. The causal tests use specific syntactic frames (“The N is a type of _____”); results may differ in other frames.

Relation to known structure. The deposit map is an instance of the low-rank OV abstraction described in prior work. Cross-position information flow has been studied through several other phenomena (IOI, binding, activation transport operators; see §6). The measurements that are, to our knowledge, not previously reported are: the cross-role comparison with proper controls for global structure (object channels stronger than the subject channel in centered R^2), the channel content beyond grammatical features (73% unnamed, with per-axis causal validation), and the layer gap between decodability and causal use of the transported copy. The finding that uncentered cosine—used in prior deposit-map work—is dominated by global positional structure may be relevant to other studies using this metric.

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at <https://github.com/EvidentSolutions/Deposit>.

A Human–AI collaborative research protocol

This research was conducted through iterative conversation with Claude (Anthropic). The human author directed research questions, validated outputs, and approved all claims. Claude conducted technical execution including code writing, experiment design, and statistical analysis. The human author takes full responsibility for all scientific claims.

B Layer-pair sweep

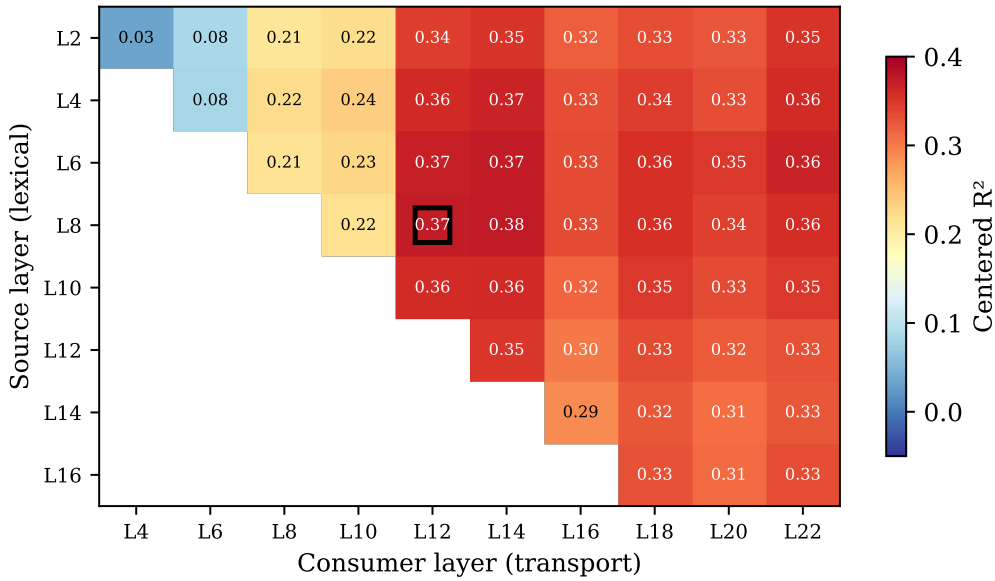


Figure 5: Centered R^2 of the deposit map (subject→verb) across source and consumer layer pairs. The black square marks L8→L12 (used throughout). The result is a broad plateau: $R^2 \approx 0.33$ – 0.38 for source layers ≤ 10 and consumer layers 12–18.

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